YEAR 11 PHYSICS, AQA GCSE, PART 3 OF 4

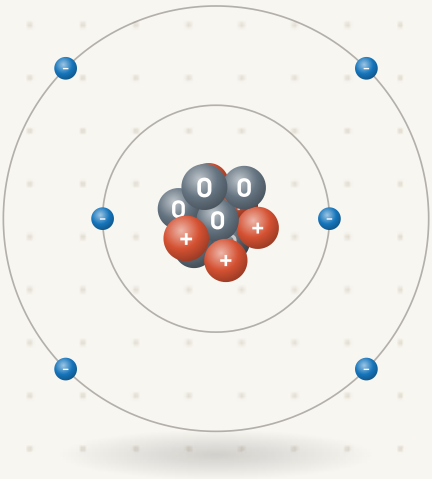
Atoms & Nuclear Radiation.

Part 3 of 4 Lessons 5–7

IN THIS BOOKLET

- Lesson 5: Radioactive decay
- Lesson 6: Nuclear equations
- Lesson 7: Properties and uses of radiation

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lessons 5–7** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try "Your turn" alone.

TRY IT
Practice that builds up, then a challenge. Quick checks from the slides follow some lessons.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this booklet highlighted)

01	Inside the atom	02	Isotopes and ions
03	From plum pudding to the nucleus	04	Energy levels and the nucleus
05	**Radioactive decay** Slides 26–30	06	**Nuclear equations** Slides 28–31
07	**Properties and uses of radiation** Slides 32–34	08	Half-life and random decay
09	Half-life calculations	10	Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

05 Radioactive decay.

AQA Physics 4.4.2.1; Combined Science 6.4.2.1

BIG QUESTION

A smoke alarm contains a radioactive source. Nothing switches it on, yet it gives out radiation for hundreds of years. Why?

BY THE END YOU CAN

- explain why some nuclei give out radiation
- describe radioactive decay as random
- define activity and count rate
- describe what alpha, beta, gamma and neutron radiation are, and how each changes the nucleus

unstable nucleus | radioactive decay | random | activity | becquerel (Bq) | count rate | Geiger–Müller tube | alpha | beta | gamma

SLIDES AND SIMULATIONS

Teaching deck, slides 26–30

Radioactive decay, activity and count rate, then alpha, beta and gamma.

panphy.app/gcsephy/decks/atoms-and-radiation/#26

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 What is an isotope?

An atom of the same element with a different number of neutrons.

2 What happens when an electron emits EM radiation?

It moves to a lower energy level, closer to the nucleus.

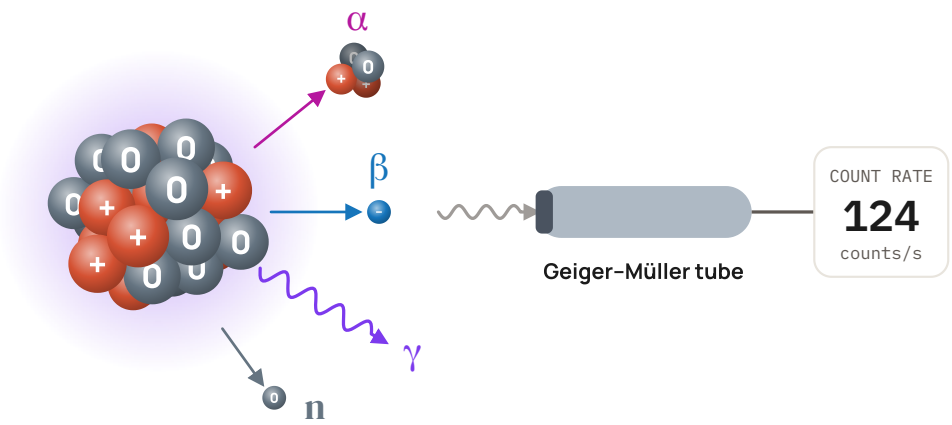
3 How many protons and neutrons are in $^{14}_{6}C$?

6 protons, 8 neutrons.

EXPLAIN

When a nucleus is unstable

Unstable nuclei



Some isotopes have an **unstable nucleus**. It gives out **radiation** as it changes to become **more stable**. This is **radioactive decay**.

Decay is **random**: you cannot predict which nucleus will decay next, or when. Nothing we do changes it: heating, cooling or chemical reactions have no effect.

Quantity	Meaning	Unit
Activity	the rate at which a source of unstable nuclei decays: the number of decays each second	becquerel (Bq); 1 Bq = 1 decay per second
Count rate	the number of decays recorded each second by a detector, such as a Geiger–Müller tube	counts per second (or per minute)

THINK OF POPCORN

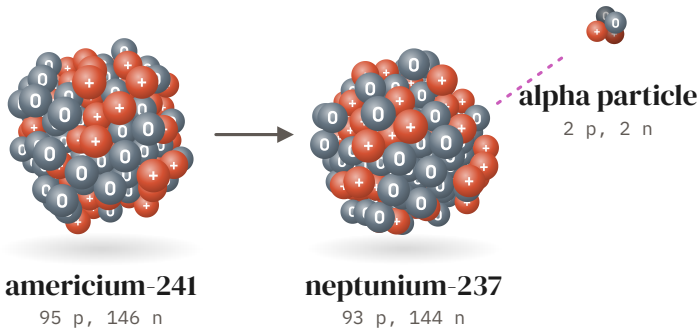
In a hot pan, you cannot tell which kernel will pop next, or when. But with lots of kernels, you can predict roughly how many will pop each minute.

Where the model breaks: heating makes popcorn pop faster, but nothing we do changes the rate of radioactive decay.

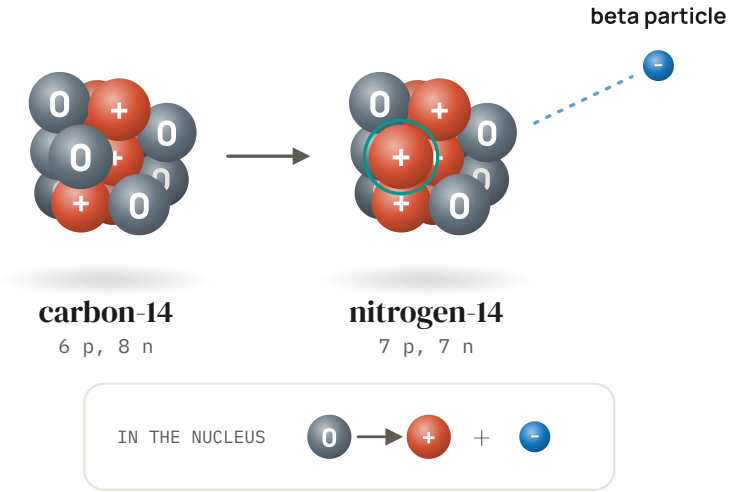
Four kinds of nuclear radiation

Radiation	What it is	Symbol	Charge	What happens in the nucleus
Alpha (α)	2 protons + 2 neutrons: a helium nucleus	$^{4}_{2}He$	+2	loses 2 protons and 2 neutrons
Beta (β)	a high-speed electron ejected from the nucleus	$^{0}_{-1}e$	−1	a neutron turns into a proton
Gamma (γ)	electromagnetic radiation from the nucleus	γ	0	loses energy only
Neutron (n)	a neutron	$^{1}_{0}n$	0	loses one neutron

Alpha decay

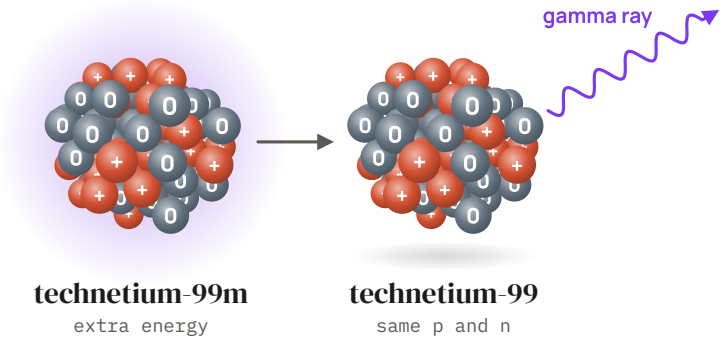


Beta decay



In beta decay, a **neutron changes into a proton**, and a fast electron is made and shoots out of the nucleus. The beta particle comes from the **nucleus**, not from the energy levels.

Gamma emission



Gamma rays are **electromagnetic waves**. The nucleus gives out extra energy, but its protons and neutrons do not change. Gamma often follows alpha or beta decay.

PRACTISE

Try it

1 Draw a line from each type of radiation to what it is.

1 alpha (α)	→ **C**	**A** a neutron
2 beta (β)	→ **B**	**B** a high-speed electron from the nucleus
3 gamma (γ)	→ **D**	**C** two protons and two neutrons
4 neutron (n)	→ **A**	**D** electromagnetic radiation

2 Which statement about radioactive decay is correct? Tick **one** box.

- ☐ Heating a source makes it decay faster.
- ☐ You can predict when a particular nucleus will decay.
- ☑ Unstable nuclei decay at random to become more stable.
- ☐ A nucleus decays when a detector is placed near it.

3 Complete the sentences.

The ___activity___ of a source is the number of decays each second. It is measured in ___becquerels___ (Bq). The number of decays a Geiger–Müller tube records each second is the ___count rate___.

WORD BANK: activity | count rate | becquerels | half-life | seconds

4 SUPPORTED Which radiation causes each change in the nucleus?

Change in the nucleus	Radiation
loses 2 protons and 2 neutrons	alpha
a neutron turns into a proton	beta
loses energy, but no particles change	gamma
loses one neutron	neutron

5 Explain what is wrong with Ava's statement.

Ava: Beta particles are electrons knocked out of the energy levels, so beta decay just turns the atom into an ion.

A beta particle comes from the nucleus: a neutron turns into a proton and a fast electron is emitted. The nucleus gains a proton, so the atom becomes a different element; it is not just an ion.

6 CHALLENGE A Geiger–Müller tube records 1800 counts in 1 minute. (a) Calculate the count rate in counts per second. (b) Suggest why the activity of the source is greater than this count rate.

(a) 1800 ÷ 60 = 30 counts per second. (b) Radiation goes out in all directions, so only some of it enters the tube; some is absorbed by the air or the tube wall before it is detected.

INVESTIGATE

Virtual lab: watch a nucleus decay

Nuclear Decay simulation, tabs 01 to 03

Open the **Nuclear Decay** simulation. For each tab, press **Decay this nucleus** and read the equation it shows.

7 Record what happens.

Tab	Parent nucleus	What leaves the nucleus?	Change in mass number	Change in atomic number	New nucleus
01 α	americium-241	alpha particle (He nucleus)	−4	−2	neptunium-237
02 β	carbon-14	beta particle (electron)	0	+1	nitrogen-14
03 γ	technetium-99m	gamma ray	0	0	technetium-99

8 Which decays make a new element? Explain using your table.

Alpha and beta: the atomic number (number of protons) changes. Gamma does not change the atomic number, so the element stays the same.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Two marks: say what the nucleus gives out *and* why.

9 Describe what is meant by radioactive decay. **[2 marks]**

An unstable nucleus gives out radiation (1) as it changes to become more stable (1). (Accept: it is random.)

10 Compare an alpha particle with a beta particle. **[3 marks]**

Alpha is 2 protons and 2 neutrons (a helium nucleus); beta is a high-speed electron (1). Alpha has a charge of +2; beta −1 (1). Alpha is much more massive / beta is made when a neutron turns into a proton (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Unstable nuclei give out radiation to become more stable: radioactive decay.
- Decay is random: we cannot predict which nucleus decays, or when.
- Activity: decays per second, in becquerel (Bq). Count rate: decays recorded per second by a detector.
- Alpha: 2p + 2n (helium nucleus). Beta: fast electron; a neutron becomes a proton. Gamma: EM radiation. Neutron: a neutron.

COVER THE PAGE AND ANSWER

1 Why do some nuclei give out radiation?

2 What is the unit of activity?

3 What is an alpha particle?

4 What happens in the nucleus in beta decay?

I CAN…

- explain why unstable nuclei decay
- describe decay as random
- define activity and count rate
- describe alpha, beta, gamma and neutron radiation

LESSON

06 Nuclear equations.

AQA Physics 4.4.2.2; Combined Science 6.4.2.2

BIG QUESTION

Radium-226 decays and becomes radon, a completely different element. How can we predict exactly which nucleus it turns into?

BY THE END YOU CAN

- use the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation using mass numbers and atomic numbers
- work out the new nucleus after alpha or beta decay
- explain the effect of each decay on the mass and charge of the nucleus

nuclear equation | parent nucleus | daughter nucleus | mass number | atomic number | balance

SLIDES AND SIMULATIONS

Teaching deck, slides 28–31

Alpha, beta and gamma equations, and the 'What changes?' table.

panphy.app/gcsephy/decks/atoms-and-radiation/#28

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 What is an alpha particle made of?

2 protons and 2 neutrons (a helium nucleus).

2 What happens in the nucleus in beta decay?

A neutron turns into a proton, and a fast electron is emitted.

3 What is the unit of activity?

Becquerel (Bq).

EXPLAIN

Balancing nuclear equations

Symbols for nuclear radiation

Emitted	Symbol	Mass number	Atomic number
Alpha particle	$^{4}_{2}\mathrm{He}$	decreases by 4	decreases by 2
Beta particle	$^{0}_{-1}\mathrm{e}$	no change	increases by 1
Gamma ray	γ	no change	no change
Neutron	$^{1}_{0}\mathrm{n}$	decreases by 1	no change

An electron has almost no mass, so its top number is 0. Its charge is −1, so its bottom number is −1. When a neutron (charge 0) turns into a proton (charge +1), the beta particle carries away the −1: the charges still balance.

! In every nuclear equation, the **top numbers add up to the same total** on both sides, and so do the **bottom numbers**. If the atomic number changes, a **new element** forms.

Writing an equation

WITH YOUR TEACHER SLIDE 28

Americium-241, $^{241}_{95}\mathrm{Am}$, emits an alpha particle and becomes neptunium (Np). Write the equation.

TOP	241 − 4 = 237
BOTTOM	95 − 2 = 93
NEW NUCLEUS	$^{237}_{93}\mathrm{Np}$
EQUATION	$^{241}_{95}\mathrm{Am} \rightarrow {}^{237}_{93}\mathrm{Np} + {}^{4}_{2}\mathrm{He}$

YOUR TURN

Thorium-230, $^{230}_{90}\mathrm{Th}$, emits an alpha particle and becomes radium (Ra). Write the equation.

TOP	230 − 4 = 226
BOTTOM	90 − 2 = 88
NEW NUCLEUS	$^{226}_{88}\mathrm{Ra}$
EQUATION	$^{230}_{90}\mathrm{Th} \rightarrow {}^{226}_{88}\mathrm{Ra} + {}^{4}_{2}\mathrm{He}$

All rights reserved.

WITH YOUR TEACHER SLIDE 29

Carbon-14, $^{14}_{6}C$, emits a beta particle and becomes nitrogen (N). Write the equation.

TOP	14 − 0 = 14
BOTTOM	6 + 1 = 7
NEW NUCLEUS	$^{14}_{7}N$
EQUATION	$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$

YOUR TURN

Strontium-90, $^{90}_{38}Sr$, emits a beta particle and becomes yttrium (Y). Write the equation.

TOP	90 − 0 = 90
BOTTOM	38 + 1 = 39
NEW NUCLEUS	$^{90}_{39}Y$
EQUATION	$^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + {}^{0}_{-1}e$

$$^{99m}_{43}Tc \rightarrow {}^{99}_{43}Tc + \gamma$$

Gamma (slide 30). The ‘m’ means the nucleus has extra energy; A and Z do not change

Finding the missing particle

WITH YOUR TEACHER SLIDE 31

Complete the equation: $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + ?$

TOP	226 − 222 = 4
BOTTOM	88 − 86 = 2
PARTICLE	$^{4}_{2}He$: an alpha particle

YOUR TURN

Complete the equation: $^{214}_{83}Bi \rightarrow {}^{214}_{84}Po + ?$

TOP	214 − 214 = 0
BOTTOM	83 − 84 = −1
PARTICLE	$^{0}_{-1}e$: a beta particle

PRACTISE

Try it

1 SUPPORTED Fill in the missing particle. Check that the top and bottom numbers balance.

$$^{239}_{94}Pu \rightarrow {}^{235}_{92}U + {}^{4}_{2}He$$

$$^{60}_{27}Co \rightarrow {}^{60}_{28}Ni + {}^{0}_{-1}e$$

2 ON YOUR OWN Fill in the missing nucleus. The element symbols are given.

$$^{222}_{86}\text{Rn} \rightarrow {}^{218}_{84}\text{Po} + {}^{4}_{2}\text{He}$$

Po = polonium

$$^{137}_{55}\text{Cs} \rightarrow {}^{137}_{56}\text{Ba} + {}^{0}_{-1}\text{e}$$

Ba = barium

$$^{5}_{2}\text{He} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$$

a helium-5 nucleus emits a neutron

3 Name the type of decay in each equation.

Equation	Type of decay
$^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\text{e}$	beta
$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$	alpha
$^{99m}_{43}\text{Tc} \rightarrow {}^{99}_{43}\text{Tc} + \gamma$	gamma

4 Complete the table to show the effect of each decay on the nucleus.

Radiation emitted	Effect on the mass of the nucleus	Effect on the charge of the nucleus
alpha	decreases (mass number – 4)	decreases (by 2)
beta	no change	increases (by 1)
gamma	no change	no change

5 Explain what is wrong with Leo's statement.

Leo: In beta decay the nucleus loses an electron, so the mass number goes down by 1.

An electron has almost no mass (0 on top), so the mass number does not change. A neutron turns into a proton, so the atomic number goes up by 1.

All rights reserved.

6 CHALLENGE Uranium-238, $^{238}_{92}U$, decays by alpha emission. The new nucleus then decays by beta emission, and the next one by beta emission again. Find the final nucleus. What do you notice?

$^{238}_{92}U \rightarrow {}^{234}_{90}Th \rightarrow {}^{234}_{91}Pa \rightarrow {}^{234}_{92}U$. The final nucleus is uranium-234: an isotope of the uranium we started with (same 92 protons, 4 fewer neutrons).

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Both numbers earn a mark. Check: top 212 = ? + 4, bottom 83 = ? + 2.

7 Bismuth-212, $^{212}_{83}Bi$, decays by alpha emission to form thallium (Tl). Complete the nuclear equation. [2 marks]

$$^{212}_{83}Bi \rightarrow {}^{208}_{81}Tl + {}^{4}_{2}He$$

Mass number 208 (1); atomic number 81 (1).

8 Technetium-99m emits gamma radiation. Explain why the element does not change. [2 marks]

Gamma radiation is electromagnetic radiation with no mass and no charge (1), so the number of protons / atomic number does not change (1).

All rights reserved.

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Symbols: alpha $^{4}_{2}\mathrm{He}$, beta $^{0}_{-1}\mathrm{e}$, gamma γ, neutron $^{1}_{0}\mathrm{n}$.
- Top numbers balance; bottom numbers balance.
- Alpha: mass number − 4, atomic number − 2.
- Beta: mass number unchanged, atomic number + 1.
- Gamma: no change to either number.

COVER THE PAGE AND ANSWER

1 What is the symbol for a beta particle?

2 What happens to the atomic number in alpha decay?

3 Which decay does not change the mass number or atomic number?

4 Complete: $^{234}_{92}\mathrm{U} \rightarrow ? + {}^{4}_{2}\mathrm{He}$

I CAN…

- write the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation
- find the new nucleus after alpha or beta decay
- explain the effect of each decay on mass and charge

LESSON

07 Properties and uses of radiation.

AQA Physics 4.4.2.1; Combined Science 6.4.2.1

BIG QUESTION

A smoke alarm uses alpha radiation, a paper mill uses beta and a hospital uses gamma. Why not use the same radiation for all three?

BY THE END YOU CAN

- compare the ionising power, range in air and penetration of alpha, beta and gamma
- explain why more ionising radiation is less penetrating
- identify radiation from absorber data
- choose a suitable source for a use and justify the choice

ionising · ion · penetration · range · absorbed · smoke alarm · thickness gauge · tracer

SLIDES AND SIMULATIONS

Teaching deck, slides 32–34

Comparing alpha, beta and gamma, penetration, and choosing a source.

panphy.app/gcsephy/decks/atoms-and-radiation/#32

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1

Complete: $^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + ?$

$^{4}_{2}He$ (an alpha particle)

2

What happens to the atomic number in beta decay?

It increases by 1.

3

What is gamma radiation?

Electromagnetic radiation from the nucleus.

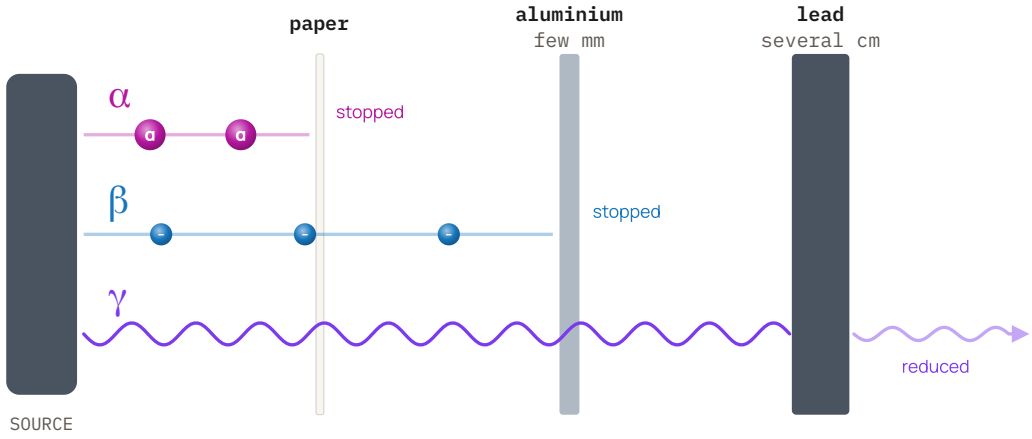
EXPLAIN

Ionising and penetrating

Nuclear radiation is ionising

As nuclear radiation passes through a material, it knocks electrons off atoms, turning them into **ions** (Lesson 2). Each ionisation takes some of the radiation's energy. Alpha particles are large with a +2 charge, so they collide with many atoms and ionise strongly: they lose their energy fast and do not travel far.

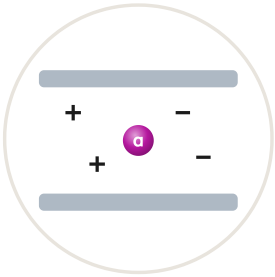
	Alpha (α)	Beta (β)	Gamma (γ)
What is it?	helium nucleus	fast electron	EM wave
Charge	+2	−1	0
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	a sheet of paper, or skin	a few mm of aluminium	reduced by thick lead or concrete



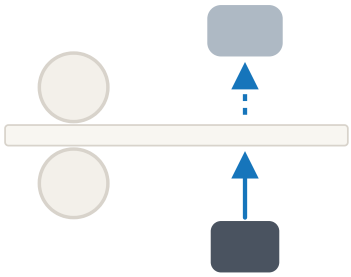
More ionising → less penetrating. Alpha loses its energy fastest, so it is stopped most easily. Gamma ionises weakly, so it travels furthest.

Atoms & Nuclear Radiation, Year 11 workbook: Part 3, Lessons 5–7 (ANSWERS), © 2026 PanPhy Labs. All rights reserved.

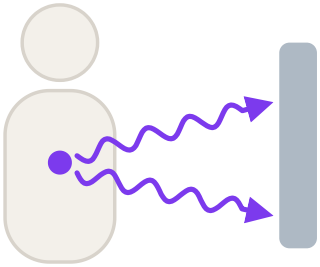
Choosing the right source



Alpha: smoke alarm
Alpha ionises the air, so a small current flows. Smoke absorbs the alpha particles, the current falls and the alarm sounds.



Beta: thickness gauge
Paper absorbs some beta. If the paper gets thicker, the count rate falls, and the rollers adjust.



Gamma: medical tracer TRIPLE ONLY
Gamma passes out of the body to a detector and is only weakly ionising.

To choose a source, think about **penetration** (must it pass through something, or be stopped?), **ionising power**, and **half-life**: how long it keeps giving out radiation (Lesson 8).

Identifying radiation from absorbers

A detector measures the count rate from a source as different absorbers are placed in between. The background count rate is 30 counts per minute.

WORKED EXAMPLE

Which radiation does this source emit?

Absorber	Count rate / counts per minute
none	900
paper	460
5 mm aluminium	455
2 cm lead	90

PAPER	900 → 460: a big fall, so alpha is present
ALUMINIUM	460 → 455: no real change, so no beta
LEAD	455 → 90: falls, but still above background (30), so gamma
ANSWER	**alpha and gamma**

YOUR TURN

Which radiation does this source emit?

Absorber	Count rate / counts per minute
none	520
paper	515
5 mm aluminium	32
2 cm lead	31

PAPER	no real change: no alpha
ALUMINIUM	falls to background: beta
LEAD	no change: no gamma
ANSWER	beta only

Atoms & Nuclear Radiation, Year 11 workbook: Part 3, Lessons 5–7 (ANSWERS), © 2026 PanPhy Labs. All rights reserved.

PRACTISE

Try it

1 SUPPORTED Complete the table.

	Alpha	Beta	Gamma
Ionising power	strong	moderate	weak
Stopped by	paper / skin	a few mm of aluminium	(reduced by) thick lead
Range in air	a few cm	about 1 m	very far

2 Which radiation is the most strongly ionising? Tick **one** box.

- [x] alpha
- [ ] beta
- [ ] gamma

3 Draw a line from each use to the radiation it needs.

1 smoke alarm	→ B	**A** gamma
2 paper thickness gauge	→ C	**B** alpha
3 medical tracer	→ A	**C** beta

4 Explain why a beta source is used to monitor the thickness of paper, and not alpha or gamma.

Alpha would be stopped completely by the paper, so the count rate would not change with thickness. Gamma would pass straight through almost unaffected. Beta is partly absorbed, so the count rate changes when the thickness changes.

5 ON YOUR OWN The background count rate is 20 counts per minute. Which radiation does this source emit? Explain using the data.

Absorber	none	paper	5 mm aluminium	2 cm lead
Count rate / counts per minute	640	380	25	21

Paper: 640 → 380, so alpha is present. Aluminium: 380 → 25, about background, so beta is present. Lead: no further change, so no gamma. The source emits alpha and beta.

All rights reserved.

6 Correct Mia's statement.

Mia: Gamma goes through everything, and nothing can stop it.

Gamma is very penetrating, but thick lead or concrete absorbs most of it: it is reduced, even if not stopped completely.

7 CHALLENGE Smoke alarms use americium-241, which has a half-life of 432 years. Suggest why a long half-life is useful here.

The source keeps giving out alpha radiation at an almost steady rate for many years, so the alarm works for its whole lifetime without replacing the source.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Link each property to the reason: ionising → loses energy quickly → short range.

8 Explain why alpha radiation is more strongly ionising but less penetrating than gamma radiation. [3 marks]

Alpha particles are large with a +2 charge, so they collide with / ionise many atoms (1). Each ionisation transfers energy, so alpha loses its energy quickly (1) and is stopped by a short distance (paper or a few cm of air). Gamma has no charge, ionises weakly, so it loses energy slowly and travels much further (1).

9 A factory uses a beta source and a detector to control the thickness of aluminium foil. Explain how the count rate shows that the foil has become too thick. [3 marks]

Thicker foil absorbs more beta radiation (1), so fewer beta particles reach the detector (1) and the count rate falls below its normal value (1).

All rights reserved.

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Nuclear radiation ionises atoms by knocking off electrons.
- Alpha: strongly ionising, a few cm in air, stopped by paper.
- Beta: moderately ionising, about 1 m in air, stopped by a few mm of aluminium.
- Gamma: weakly ionising, very far in air, reduced by thick lead or concrete.
- More ionising → less penetrating. Choose a source by penetration, ionising power and half-life.

COVER THE PAGE AND ANSWER

1 What does ionising mean?

2 What stops beta radiation?

3 Which radiation is used in a smoke alarm?

4 Why is gamma used as a medical tracer?

I CAN…

- compare the properties of alpha, beta and gamma
- explain the link between ionising and penetration
- identify radiation from absorber data
- choose and justify a source for a use

All rights reserved.

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius $\approx 1 \times 10^{-10}$ m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

REFERENCE

Key words from this booklet.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Activity: The rate at which a source of unstable nuclei decays. Unit: becquerel (Bq).

Alpha particle (α): Two protons and two neutrons: a helium nucleus. Strongly ionising; stopped by paper.

Becquerel (Bq): Unit of activity: 1 Bq = 1 decay per second.

Beta particle (β): A high-speed electron ejected from the nucleus when a neutron turns into a proton.

Count rate: The number of decays recorded each second by a detector.

Daughter nucleus: The new nucleus formed when a nucleus decays.

Gamma ray (γ): Electromagnetic radiation emitted from a nucleus. Weakly ionising; very penetrating.

Ionising: Able to knock electrons off atoms, turning them into ions.

Nuclear equation: An equation showing a decay, in which the mass numbers and atomic numbers balance.

Penetration: How far radiation can travel through materials before it is absorbed.

Radioactive decay: An unstable nucleus giving out radiation to become more stable. It is random.

Unstable nucleus: A nucleus that will decay and give out radiation.

All rights reserved.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 5: RADIOACTIVE DECAY

DO NOW

1. An atom of the same element with a different number of neutrons.
2. It moves to a lower energy level, closer to the nucleus.
3. 6 protons, 8 neutrons.

COVER THE PAGE

1. They are unstable; they decay to become more stable.
2. Becquerel (Bq).
3. 2 protons and 2 neutrons (a helium nucleus).
4. A neutron turns into a proton (and a fast electron is emitted).

LESSON 6: NUCLEAR EQUATIONS

DO NOW

1. 2 protons and 2 neutrons (a helium nucleus).
2. A neutron turns into a proton, and a fast electron is emitted.
3. Becquerel (Bq).

COVER THE PAGE

1. $^{0}_{-1}e$
2. It decreases by 2.
3. Gamma emission.
4. $^{230}_{90}Th$

LESSON 7: PROPERTIES AND USES OF RADIATION

DO NOW

1. $^{4}_{2}He$ (an alpha particle)
2. It increases by 1.
3. Electromagnetic radiation from the nucleus.

COVER THE PAGE

1. Knocking electrons off atoms, making ions.
2. A few mm of aluminium.
3. Alpha.
4. It passes out of the body to a detector and is only weakly ionising.

All rights reserved.